

Maxwell Glynn Wainwright

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EDUCATION

Worcester Polytechnic Institute (WPI), Worcester MA

Expected Graduation: 2028

- Bachelor of Science in both **Computer Science** and **Interactive Media and Game Design (IMGD)**
- Cumulative GPA: **3.66 / 4.0**

EXPERIENCE

iOS Software Engineer | David Music LLC

2026 – Present

- Design and deliver production software end to end in a **60,000-line object-oriented** codebase, applying **OOP** concepts including **classes, inheritance, polymorphism, encapsulation, and interfaces**.
- Integrate **cloud-hosted backend services** on **Google Cloud (Firebase/Firestore)**, implementing **real-time data synchronization** and offline caching behind interface-driven service abstractions.
- Adhere to **quality standards** through strict **test-driven development** and **peer code review**, maintaining a **26,000-line automated test suite** across a five-engineer team.
- **Collaborate with project team members** to research, design, and refactor **core architecture**, enhancing the **end-to-end user experience** across scheduling, customer, and account management modules.

Software Engineer Intern | Hunu Interactive

2025 – Present

- Developed a highly scalable **CSV-to-ScriptableObject** data pipeline, optimizing designer workflows.
- Engineered a robust **minimax AI** opponent, providing designers with **non-destructive parameter tuning**.
- Architected a **cross-platform multiplayer** lobby utilizing **Unity Relay** with **server-side validation**.

Game Developer: G-Force Studios

2024

- **Collaborated** directly with **industry veterans** to prototype a **mobile application** in **Unity (C#)**, engineering core systems with strict **object-oriented principles** to deliver a **minimum viable product**.

SKILLS

- **Languages:** Java, Python, C++, JavaScript, TypeScript, Swift, C#, SQL, HTML/CSS, Mandarin
- **Frameworks & Libraries:** React, Node.js, Express, PyTorch, CoreML, UIKit, SwiftUI, Tailwind CSS, Prisma ORM, SFML
- **Cloud & Data:** Google Cloud (Firebase/Firestore), PostgreSQL, SQLite, REST APIs, Client-Server Architecture
- **Concepts & Tools:** Object-Oriented Programming, Generative AI, Transformers, Machine Learning, Test-Driven Development, Agile, Code Review, Git, Unity

PROJECTS

Hanover Insurance Web Application Prototype | PERN Stack, TypeScript, Agile

2026

- I worked on a **10-person team** to apply **Agile** development methodologies in the creation of a web application prototype running the cloud for **Hanover Insurance**.
- As a **Team Lead software engineer**, I was the **technical leader** of the **frontend** subgroup.
- I was responsible for writing the Main **Content Management** and **User Management components** of the application.
- The web application was developed using the **PERN** (PostgreSQL, Express, React, Node.js) stack with Typescript, Prisma ORM, and tailwind.

DoomClimb | Generative AI, Swift, PyTorch, CoreML, CoreBluetooth, SQLite

2026

- Developed and deployed a **native iOS App Store** application powered by a **custom-trained generative AI model**, a **2M-parameter GPT-style transformer** that **autoregressively generates** novel bouldering routes.
- Engineered an end-to-end **PyTorch machine learning pipeline** trained on **60,000 SQLite** records, exported to **CoreML** for robust **on-device model inference** independent of cloud services.
- Designed a **graph-reachability algorithm** to validate model output and reverse-engineered a **BLE protocol** with **custom packet chunking** to drive **476** motorized hardware LEDs.

Elementals | C++, SFML 3.0

2025

- Built a **custom 2D engine** from scratch in **C++**, implementing **object-oriented managers** for display, hardware input, logging, and state management.
- Developed **tree data structures** to parse input sequences and engineered **deterministic state management** with **custom physics** and **collision resolution** for **local multiplayer**.